

Anda Denić

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Skills

- AI, ML, Machine Learning, Data Science, Data Analytics, Computer Vision, NLP
- **Python**: NumPy, Scikit-learn, Matplotlib, PyTorch, Pandas, SciPy, Gymnasium
- **Mathematics**: Probability, Statistics, Multivariable Calculus, Optimization Theory, Linear Algebra
- **English**

Education

[My Academic Transcript](#)

PhD student in Applied Mathematics

[Faculty of Sciences and Mathematics](#), University of Nis, Serbia

- Focus on **statistics** and **machine learning**

MSc in Machine Learning and Artificial Intelligence

[Faculty of Sciences and Mathematics](#), University of Nis, Serbia

- Defended master's thesis "Image captioning using ResNet and LSTM models"
- GPA: 9.88/10.0

BSc in Mathematics

[Faculty of Sciences and Mathematics](#), University of Nis, Serbia

- GPA: 9.16/10.0

Gymnasium "Svetozar Marković", Nis, Serbia

- [Class for students talented in mathematics](#)

Experience

Teaching Associate

[Faculty of Electronic Engineering](#), University of Nis

February 2024 - present

- Calculus, Mathematical Analysis, Numerical Analysis, Probability and Statistics

Machine Learning Engineer Internship

[Diffine](#)

April 2023 – Jun 2023.

- Machine Learning, Computer Vision, Data Analysis

Other

Studygroup Mathematics with Industry 2025

January 2025, Utrecht University, Utrecht, Netherlands

- [SWI](#) is a combined industrial–academic workshop where mathematicians work in groups on problems submitted by companies.
- My group and I worked for a week on a [problem with implementing Bayesian marketing mix modelling](#) given by pharmaceutical firm Opella.
- Our results are published as **scientific proceeding** [“Tackling Multicollinearity in Marketing Mix Models: A Bayesian Hierarchical Shrinkage Approach”](#).

Conference talk

May 2024, University of Kragujevac, Serbia

- I gave a conference talk in [The Third Serbian International Conference on Applied Artificial Intelligence](#)
- I presented my work in bioinformatics ["Exploring Word2Vec Models for Capturing the Similarity of Codon Embeddings"](#)
- I received the **award for the best student work**
- [Video recording of that conference talk](#)

STEM Lecturer

American Corner Niš, Serbia

April 2025 – Jun 2025.

- I designed and delivered 10 STEM workshops in English on fundamental concepts, problems, and algorithms in statistics, machine learning, data science, and neural networks.

Machine Learning and AI projects

Image Captioning

<https://github.com/andjadenic/ImageCaptioning>

- Technologies: **Python, PyTorch, torchvision, pandas**
- **Computer Vision, NLP, Encoder Decoder, LSTM, ResNet**
- For a given image that machine sees for the first time as input, it should generate a sentence description of that picture as output.
- The project follows an encoder-decoder structure where ResNet is encoder and LSTM is decoder.
- Training data is [MC COCO](#) dataset that includes 5 captions for 330k images.

Sentiment Analysis

<https://github.com/andjadenic/Sentiment-Analysis>

- Technologies: **Python, PyTorch, genism, pandas**
- **Supervised Learning, NLP, LSTM**
- The goal of this project is to enable machine to **automatically classify a movie review** as either "positive" or "negative" based on a given dataset of review-sentiment pairs.
- The model is a multilayer LSTM with an additional fully connected linear layer and sigmoid activation function that produces prediction.
- Model is trained on "IMDB Review Dataset" that contains 50,000 movie reviews labeled as 'positive' or 'negative'.
- The model with tuned hyperparameters achieved 88.93% accuracy on test dataset.

Neural Hidden Markov Model

<https://github.com/andjadenic/Neural-Hidden-Markov-Model>

- Technologies: **Python, PyTorch, pandas**
- **CNN, Hidden Markov Model, LSTM, POS tagging**
- The goal of this project is to automate **part-of-speech tagging**, the process of identifying a word in a text as a noun, verb, adjective, adverb, etc., based on both its definition and its context.
- Hidden Markov Model is assumed to be the base underlying model and transition and emission matrices are neutralized using LSTM and CNN.
- Inference is done using the Viterbi algorithm that finds the most probable hidden states (tags) for given observations (sentence).

Exploring Word2Vec Models for Capturing the Similarity of Codon Embeddings

github.com/andjadenic/Word2Vec-models-for-codon-embeddings

- Technologies: **Python, Matplotlib, SciPy, Gensim**
- **Bioinformatics, Machine Learning, NLP, Sequential Data**
- Project empirically analyzes the quality of 3-mer (codon) [word2vec](#) embeddings of the genome sequences of [V. Cholerae](#) and [E. Coli](#).

Reinforcement Learning Algorithms Using Gymnasium Frozen Lake

github.com/andjadenic/RL_frozen_lake

- Technologies: **Python, Gymnasium**
- Bellman Equation, Policy Evaluation and Improvement, Monte Carlo Methods, Temporal Difference (SARSA, Expected SARSA, Q-Learning)
- Project implements all basic RL algorithms from scratch using the popular Toy Text environment in Gymnasium Frozen Lake.
- I wrote supplementary [notes](#) that explain all the basic methods.

Bayesian Linear Regression Inference: Ironman Triathlon World Championship Data

github.com/andjadenic/BayesianRegression

- Technologies: **Python, Pandas, NumPy, SciPy, Matplotlib**
- **Bayesian Inference, Linear Regression**
- Project explores the usage of Bayesian linear regression inference and running time as a predictor of the overall finish time of participants in the Ironman Triathlon World Championship.